Intro to Web Development with Flask

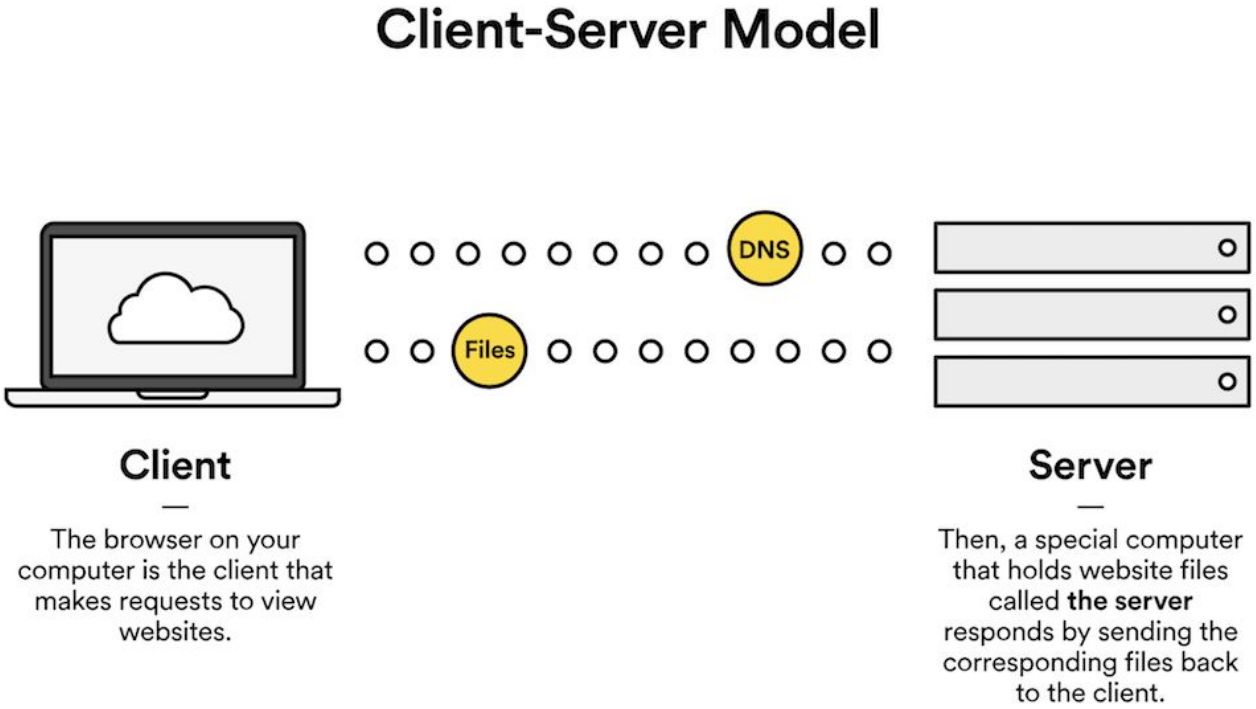
GENERAL ASSEMBLY

Agenda

- The client/server relationship model
- Front-end vs Back-end web development
- Web Frameworks
- Creating a (very, very) simple Flask app from scratch!

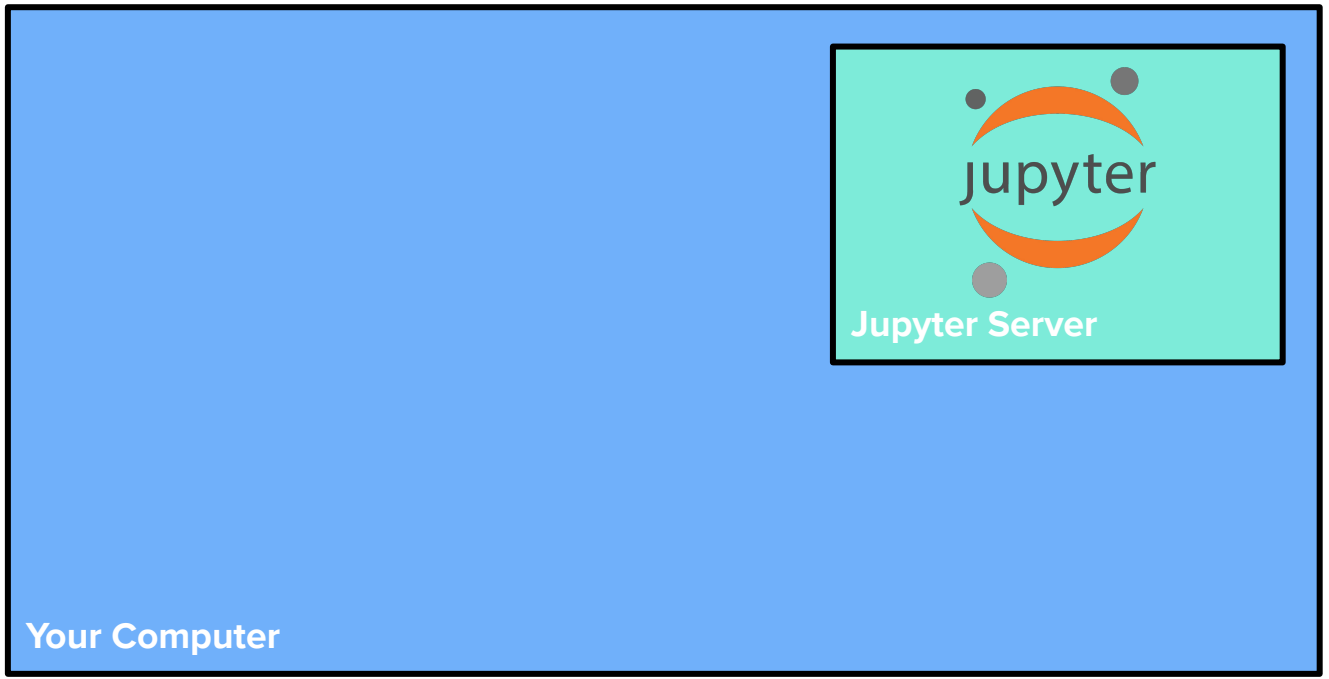
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The Client-Server Model

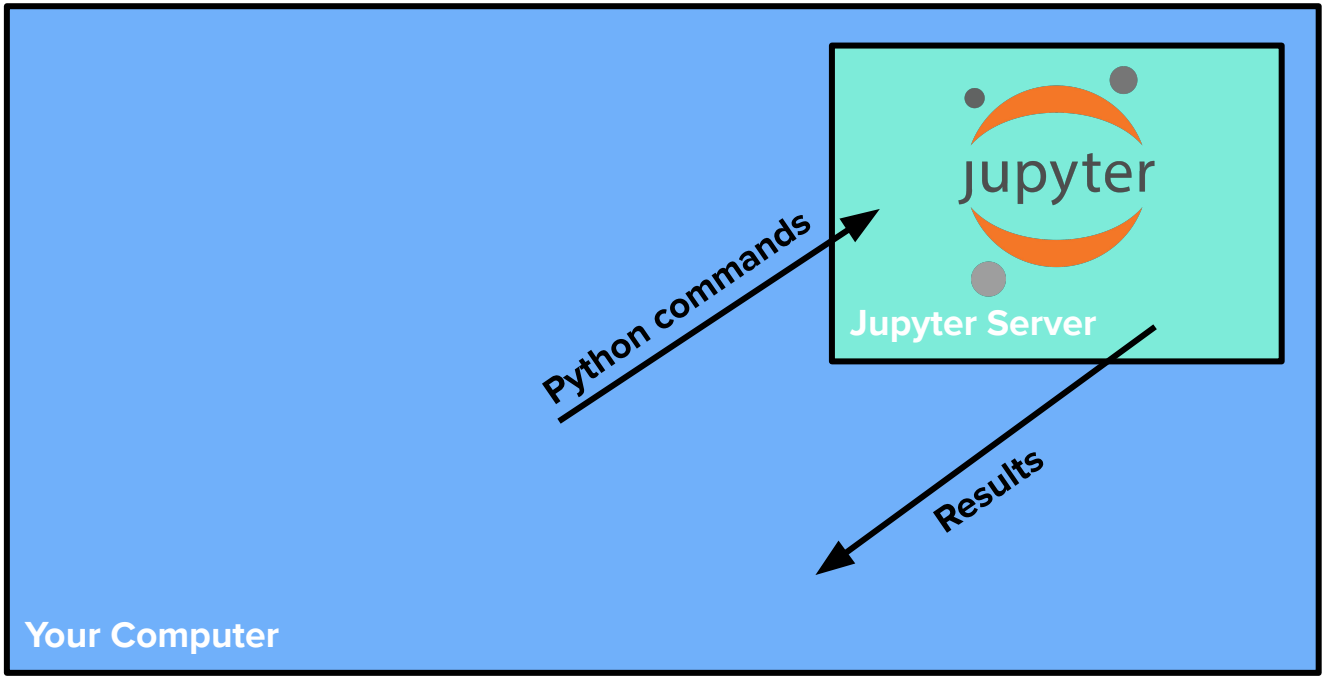


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We've actually be doing this for months!

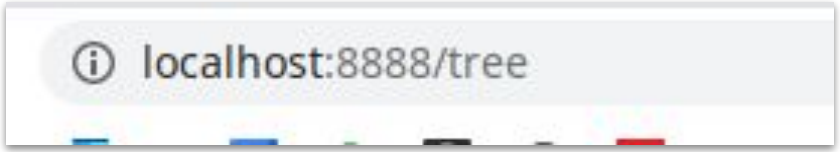


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The “Local Host”



localhost is an abbreviation for the IP address **127.0.0.1**, which is how your computer refers to itself!

8888 is the **port number** that is **listening** for your HTTP requests.

/tree is the **endpoint** for your HTTP request.

Ports

If your computer was an apartment building, its **ports** would be the individual mailboxes for the different housing units.

Ports 1 through 1024 are typically reserved for internal processes on your computer. The rest are a free-for-all.



https://i.pinimg.com/originals/6c/44/45/6c4445ac3bf6cd7751f5cda3f02a06c3.jpg

Flask On The Web (Oversimplified!)

By default, Flask uses port **5000**. So when you go to **pinterest.com** (which uses Flask!), the following occurs:

- Your web browser sends an HTTP request: **GET pinterest.com**
- This request gets routed to a **Domain Name System (DNS) Server**, which routes your request as: **GET 151.101.64.84**
- Port 5000 hears your requests, processes it, and sends the response back.

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Today's Lesson on localhost

We won't be hosting web apps on the web today. We'll be running **Flask apps locally**. That is, our Flask server will be running on:

localhost:5000

aka

127.0.0.1:5000

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Front-end vs Back-end



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Front-end vs Back-end

The **front-end** of a web page consists of all the visuals. HTML placement, styling, etc. The front-end is built using HTML, CSS, and usually Javascript.

The **back-end** of a web page consists of everything else. Handling server requests, database transactions, doing any sort of calculations, etc. The back-end is built in your favorite programming language.

Web Frameworks

Web frameworks are programming libraries that make coding the client-server relationship much easier. Most programming languages have a few to choose from:

Web Frameworks in Various Languages







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Web Frameworks in Various Languages











Sinatra



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Web Frameworks in Various Languages





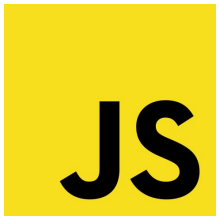






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(These are all front-end, but back-end JS frameworks do exist)

Let's Get Started!



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